A2TTA Knockout Ablation

Knockout row	Method	Calibrator	Change
Ours (full)	tta_ctx_local	film	FiLM + online TTA + local clone
− local clone	tta_all	film	remove local clone
− FiLM → affine	tta_ctx_local	affine	replace FiLM with static affine
− online TTA	calibrator	film	disable online updates
− calibration stack → backbone	backbone	—	frozen backbone only

PEMS03

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	11.18±0.03	12.10±0.04	13.74±0.07	**12.19±0.04**	—
− local clone	11.21±0.03	12.16±0.04	13.91±0.06	12.26±0.04	+0.07
− FiLM → affine	11.47±0.05	12.48±0.06	14.42±0.09	12.61±0.06	+0.42
− online TTA	11.19±0.03	12.13±0.04	13.85±0.06	12.22±0.04	+0.04
− calibration stack → backbor	11.76±0.09	12.75±0.09	14.67±0.12	12.89±0.09	+0.70

PEMS04

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	16.56±0.04	18.01±0.04	20.62±0.08	**18.15±0.05**	—
− local clone	16.63±0.04	18.15±0.06	20.96±0.12	18.32±0.07	+0.16
− FiLM → affine	16.99±0.07	18.65±0.10	21.86±0.19	18.88±0.11	+0.72
− online TTA	16.58±0.03	18.08±0.04	20.85±0.08	18.24±0.04	+0.09
− calibration stack → backbon	17.41±0.14	19.05±0.14	22.23±0.17	19.27±0.15	+1.12

PEMS05

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	9.63±0.01	10.21±0.03	11.28±0.05	**10.27±0.03**	—
− local clone	9.71±0.02	10.32±0.03	11.46±0.06	10.39±0.03	+0.12
− FiLM → affine	9.92±0.04	10.58±0.05	11.88±0.08	10.67±0.05	+0.41
− online TTA	9.70±0.01	10.31±0.03	11.45±0.05	10.38±0.03	+0.11
− calibration stack → backbor	10.11±0.08	10.75±0.09	12.04±0.12	10.85±0.09	+0.58

PEMS06

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	11.50±0.14	12.31±0.16	13.77±0.18	**12.38±0.15**	—
− local clone	11.64±0.15	12.53±0.21	14.22±0.30	12.63±0.21	+0.25
− FiLM → affine	11.83±0.22	12.72±0.25	14.42±0.31	12.82±0.25	+0.44
− online TTA	11.50±0.10	12.33±0.14	13.87±0.18	12.41±0.13	+0.03
− calibration stack → backbor	12.49±0.58	13.31±0.55	14.89±0.54	13.40±0.55	+1.02

PEMS03

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	10.57±0.03	11.08±0.03	11.93±0.05	**11.10±0.04**	—
− local clone	10.61±0.03	11.14±0.03	12.01±0.06	11.16±0.04	+0.06
− FiLM → affine	10.96±0.09	11.47±0.08	12.33±0.09	11.51±0.09	+0.40
− online TTA	10.64±0.03	11.18±0.04	12.05±0.06	11.20±0.04	+0.09
− calibration stack → backbor	11.23±0.14	11.72±0.12	12.57±0.13	11.76±0.13	+0.66

PEMS04

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	15.81±0.07	16.65±0.11	18.06±0.18	**16.70±0.11**	—
− local clone	15.94±0.10	16.81±0.13	18.28±0.22	16.86±0.14	+0.16
− FiLM → affine	17.32±0.91	18.14±0.92	19.54±0.96	18.20±0.92	+1.50
− online TTA	15.91±0.08	16.78±0.11	18.24±0.20	16.82±0.12	+0.13
− calibration stack → backbon	17.78±0.76	18.50±0.81	19.84±0.90	18.57±0.82	+1.87

PEMS05

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	9.20±0.02	9.50±0.02	10.01±0.04	**9.52±0.02**	—
− local clone	9.25±0.02	9.55±0.02	10.07±0.04	9.57±0.03	+0.05
− FiLM → affine	9.57±0.04	9.84±0.03	10.35±0.04	9.87±0.04	+0.35
− online TTA	9.28±0.01	9.59±0.01	10.11±0.03	9.61±0.02	+0.09
− calibration stack → backbor	9.75±0.04	10.01±0.03	10.51±0.03	10.05±0.03	+0.52

PEMS06

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	11.73±0.57	12.15±0.58	12.88±0.58	**12.18±0.58**	—
− local clone	12.25±0.56	12.68±0.58	13.42±0.63	12.70±0.59	+0.53
− FiLM → affine	12.06±0.47	12.43±0.46	13.11±0.45	12.47±0.46	+0.29
− online TTA	11.00±0.21	11.43±0.22	12.18±0.24	11.46±0.22	-0.72
− calibration stack → backbon	12.33±0.46	12.67±0.45	13.30±0.44	12.70±0.45	+0.53